1. Linear Equations in Two Variables

Q1. Solve by elimination: $5x - 3y = 7$ and $3x + 2y = 1$.
Given: $5x - 3y = 7$ ...(1), $3x + 2y = 1$ ...(2)
Multiply (1) by 2 and (2) by 3 to eliminate y:
$2\times(1)$: $10x - 6y = 14$ ...(3)
$3\times(2)$: $9x + 6y = 3$ ...(4)
Add (3) and (4): $19x = 17 \Rightarrow x = 17/19$
Substitute x in (2): $3(17/19) + 2y = 1$
$51/19 + 2y = 19/19 \Rightarrow 2y = -32/19 \Rightarrow y = -16/19$
Verification: Put $x = 17/19$, $y = -16/19$ in both equations; both satisfy.
Answer: $x = 17/19$, $y = -16/19$.

Q2. Solve by substitution: $x + 2y = 4$ and $3x - y = 5$.
Given: $x + 2y = 4$ ...(1), $3x - y = 5$ ...(2)
From (1): $x = 4 - 2y$
Substitute in (2): $3(4 - 2y) - y = 5$
$12 - 6y - y = 5 \Rightarrow -7y = -7 \Rightarrow y = 1$
Then $x = 4 - 2(1) = 2$
Answer: $x = 2$, $y = 1$.

Q3. Find the point of intersection of lines $2x + 3y = 6$ and $4x - 3y = 12$.
Given: $2x + 3y = 6$ ...(1), $4x - 3y = 12$ ...(2)
Add (1) and (2): $6x = 18 \Rightarrow x = 3$
Put $x = 3$ in (1): $2(3) + 3y = 6 \Rightarrow 6 + 3y = 6 \Rightarrow y = 0$
Answer: Point of intersection = (3, 0).

Q4. Form a pair of linear equations and solve: The sum of two numbers is 27 and their difference is 5.
Let the numbers be x and y ($x > y$).
Sum: $x + y = 27$ ...(1)
Difference: $x - y = 5$ ...(2)
Add (1) and (2): $2x = 32 \Rightarrow x = 16$
Then $y = 27 - 16 = 11$
Answer: The numbers are 16 and 11.

Q5. A two-digit number is such that the sum of its digits is 11. If we interchange the digits, the new number is 9 less than twice the original number. Form the pair of linear equations and find the number.
Let the tens digit = x and units digit = y.
Then $x + y = 11$ ...(1)
Original number = $10x + y$
Interchanged number = $10y + x$
Given condition: $10y + x = 2(10x + y) - 9$
$\Rightarrow 10y + x = 20x + 2y - 9$
$\Rightarrow 8y - 19x + 9 = 0 \Rightarrow 19x - 8y = 9$ ...(2)
From (1): $y = 11 - x$
Put in (2): $19x - 8(11 - x) = 9 \Rightarrow 19x - 88 + 8x = 9$
$\Rightarrow 27x = 97$
$\Rightarrow x = 97/27$, $y = 11 - 97/27 = 200/27$
Original number = $10x + y = 10(97/27) + 200/27 = 130/3$
Answer (as per given data): Number = 130/3.
Board Note: Since digits of a two-digit number must be whole numbers 0–9, the given numerical data appears inconsistent/misprinted. Method and algebraic result are shown correctly.

Q6. The sum of the present ages of a father and his son is 50 years. Ten years ago, the father's age was three times the son's age. Find their present ages.
Let present ages be Father = F years, Son = S years.
$F + S = 50$ ...(1)
Ten years ago: $F - 10 = 3(S - 10)$
$\Rightarrow F - 10 = 3S - 30 \Rightarrow F = 3S - 20$ ...(2)
Put (2) in (1): $(3S - 20) + S = 50 \Rightarrow 4S = 70 \Rightarrow S = 17.5$
Then $F = 50 - 17.5 = 32.5$
Answer: Son = 17.5 years (17 years 6 months), Father = 32.5 years (32 years 6 months).

Q7. Draw the graph of $2x - 3y = 6$ and $x + y = 4$. Hence find the coordinates of their point of intersection and verify it algebraically.
Line 1: $2x - 3y = 6$
Two points: If $y = 0 \Rightarrow x = 3 \Rightarrow (3, 0)$; If $x = 0 \Rightarrow y = -2 \Rightarrow (0, -2)$
Line 2: $x + y = 4$
Two points: If $y = 0 \Rightarrow x = 4 \Rightarrow (4, 0)$; If $x = 0 \Rightarrow y = 4 \Rightarrow (0, 4)$
Plot these points, draw both lines, and mark their intersection.
Algebraic verification: From $x + y = 4 \Rightarrow y = 4 - x$
Substitute in $2x - 3y = 6$: $2x - 3(4 - x) = 6$
$\Rightarrow 2x - 12 + 3x = 6 \Rightarrow 5x = 18 \Rightarrow x = 18/5$
$y = 4 - 18/5 = 2/5$
Answer: Point of intersection = (18/5, 2/5).

Q8. For what value of k will the following pair have (i) no solution (ii) infinitely many solutions? $3x - 5y = 7$, $6x - 10y = k$.

Given: $3x - 5y = 7$ ...(1), $6x - 10y = k$ ...(2)
Compare ratios: $a1/a2 = 3/6 = 1/2$ and $b1/b2 = (-5)/(-10) = 1/2$
Thus $a1/a2 = b1/b2$.
(i) Infinitely many solutions when $a1/a2 = b1/b2 = c1/c2$
$\Rightarrow 7/k = 1/2 \Rightarrow k = 14$
(ii) No solution when $a1/a2 = b1/b2 \neq c1/c2 \Rightarrow k \neq 14$
Answer: Infinitely many solutions for $k = 14$; No solution for $k \neq 14$.

Q9. A shop sells pencils at ₹5 each and pens at ₹12 each. In one day, the shopkeeper sold 40 items and collected ₹320. Form the pair of linear equations and find how many pencils and pens were sold.
Let pencils = x and pens = y.
Total items: $x + y = 40$ ...(1)
Total amount: $5x + 12y = 320$ ...(2)
From (1): $x = 40 - y$
Put in (2): $5(40 - y) + 12y = 320$
$200 - 5y + 12y = 320 \Rightarrow 7y = 120 \Rightarrow y = 120/7$
$x = 40 - 120/7 = 160/7$
Answer (as per given data): Pencils = 160/7, Pens = 120/7.
Board Note: Number of items must be whole numbers. Since solution is not integral, the given total amount appears inconsistent/misprinted. Method and algebraic result are correct.

Q10. Determine whether the lines represented by $4x + 6y = 12$ and $2x + 3y = 8$ are intersecting, parallel or coincident. Justify your answer.
Given: $4x + 6y = 12$ ...(1), $2x + 3y = 8$ ...(2)
Ratios: $a1/a2 = 4/2 = 2$, $b1/b2 = 6/3 = 2$, $c1/c2 = 12/8 = 3/2$
Since $a1/a2 = b1/b2$ but $a1/a2 \neq c1/c2$, the lines are parallel and distinct.
Answer: The lines are parallel (no solution).